

AI Career Intelligence Platform

Database Design Document

ER Diagram • Database Schema • Migration Strategy

Version: 1.0

Prepared By: Akshay Mohan

Table of Contents

- 1. Database Overview
- 2. ER Diagram
- 3. Database Schema
- 4. Migration Strategy
- 5. Conclusion

1. Database Overview

The platform uses PostgreSQL as the primary relational database. Tables are normalized to reduce redundancy and support resume analysis, job matching, AI recommendations, and user management.

2. ER Diagram (Conceptual)

```
Users
|
+--< Resumes >--< ResumeSkills >-- Skills
|
+--< JobMatches >-- Jobs
|
+--< LearningRoadmaps
|
+--< ChatHistory
```

3. Database Schema

Table	Purpose
Users	Stores user accounts
Resumes	Uploaded resumes
Skills	Master skill list
ResumeSkills	Maps resumes to skills
Jobs	Available job listings
JobMatches	Stores recommendation results
LearningRoadmaps	Personalized learning plans
ChatHistory	AI assistant conversations

Primary Keys: Each table uses an ID column.

Foreign Keys: user_id, resume_id, skill_id and job_id maintain relationships.

4. Migration Strategy

- Use Alembic with SQLAlchemy for schema migrations.
- Maintain version-controlled migration scripts in Git.
- Apply migrations automatically during deployment.
- Always back up the database before production upgrades.
- Support rollback to the previous stable migration if required.

5. Conclusion

This document provides the initial database design for Version 1.0 of the AI Career Intelligence Platform. It establishes the core entities, relationships, schema structure, and migration approach for future development.